SERVER



Architecting Openness: How Switches Enable the Future of Open Adapter Card Platforms

Chris Blackburn

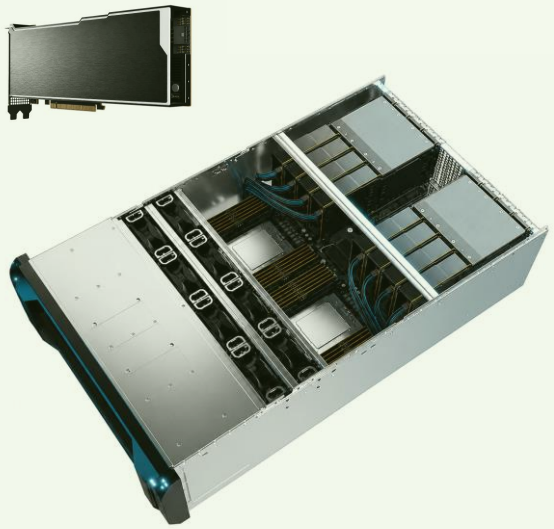
Director of Platform Solutions Architecture, Astera Labs

OCP PLATINUM® Member

AI Architectures: Pick Your Size (S,M,L,XL)

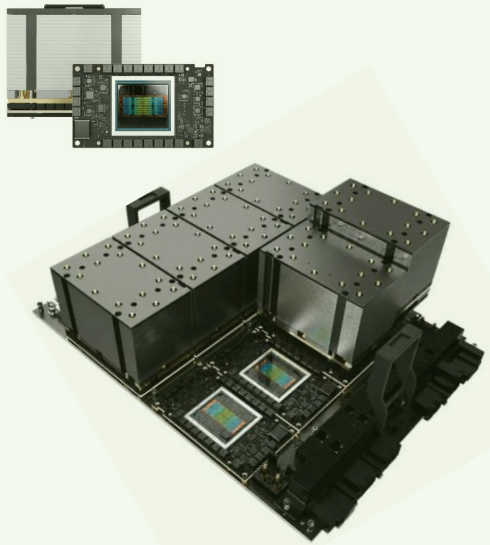
S

Add-In-Cards & Servers



M

OAM & Servers



L

Rack Scale

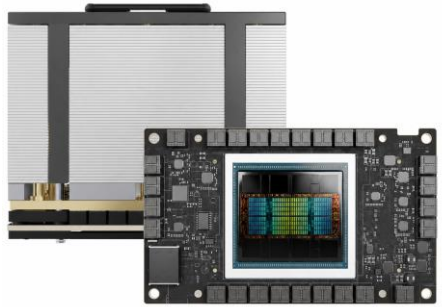
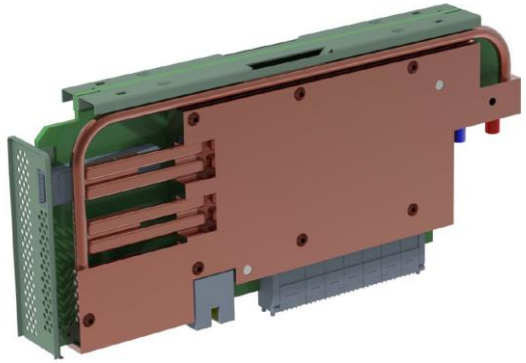


XL

Cluster Scale



Filling Gaps: Enter Open Adapter Card (OAC)

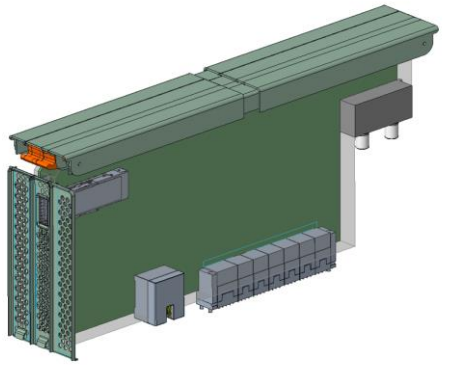
	Add-In-Card (AIC)	Open Accelerator Module (OAM)	Open Adapter Card (OAC)
			
Orientation	Vertical	Horizontal	Vertical
Max Power	600W	700W	5,000W*
Cooling	Air, Adapted DLC	Air & DLC	Air & DLC
PCB Area	FHFL: 346cm^2	168cm^2	598cm^2
Connector Interface	CEM Gold Finger Card Edge	Mezzanine Connectors	Blind Mate Backplane
Scale-Up Lanes	None dedicated	Up to x112 (7 x16)	Up to x112*

*OAC specification still under development and may not be explicit on these numbers

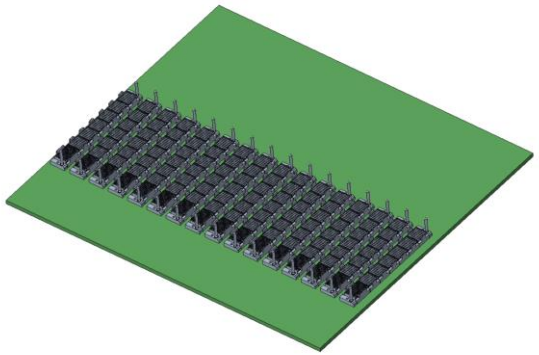
OAC Platform Architecture

- OAC is an open, modular accelerator card specification defined by OCP
- 8 or 16 OAC modules in a system
- Systems include compute, storage, networking, and management devices
- Key interfaces:
  - Scale-Up Network: UALink™, ESUN/Ethernet, Proprietary
  - PCIe®: XPUs, CPUs, Scale-Out NICs, SSDs, & BMC

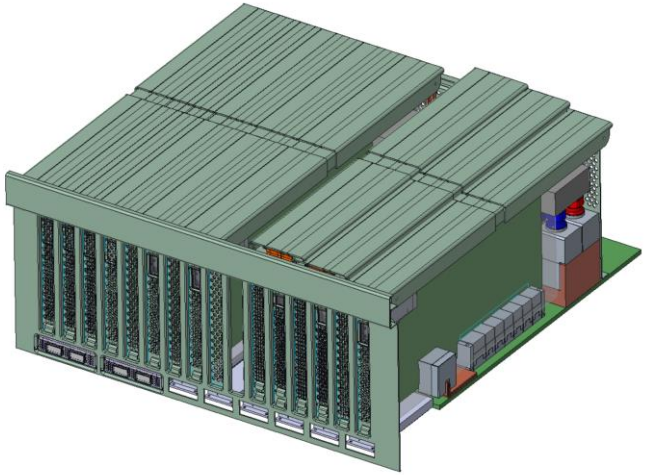
OAC Module



OAC Baseboard



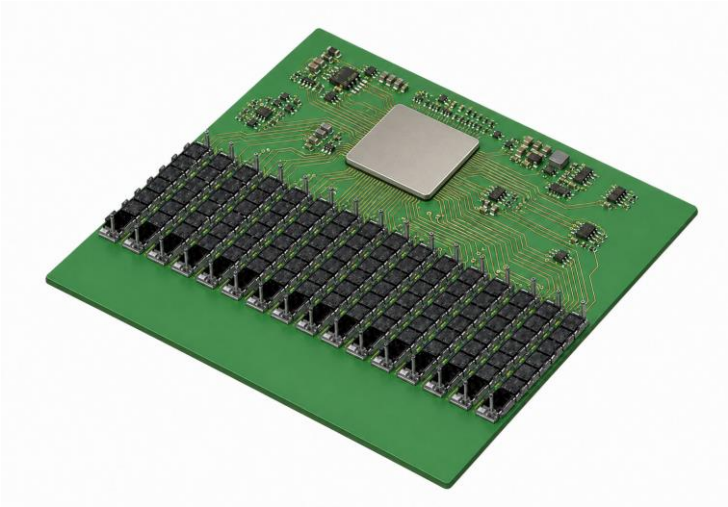
OAC System



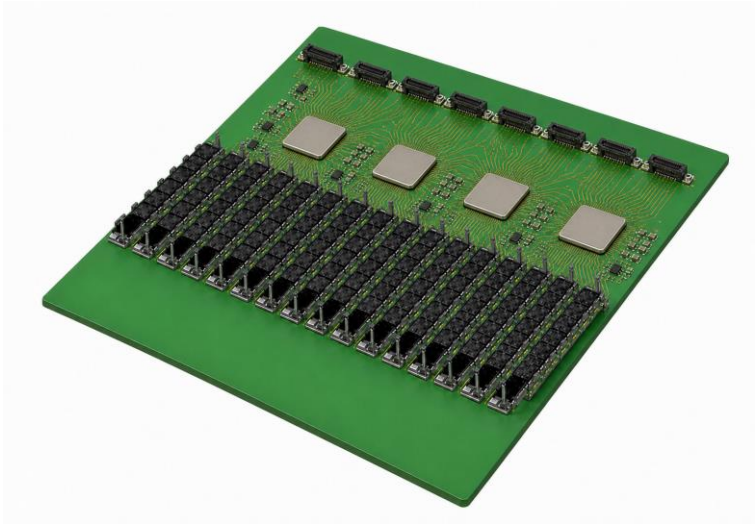
"With so many components in a modern AI system — how do they all talk to each other?"

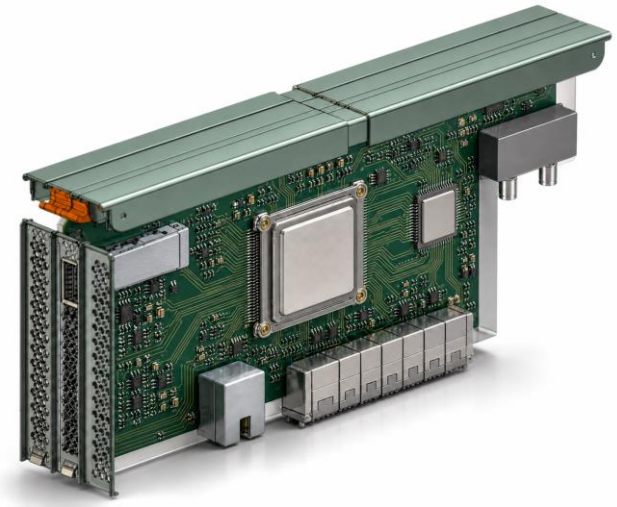
Switches Are Critical to OAC Platform Topology

Scale-Up Switches for XPU Scale-Up Fabric



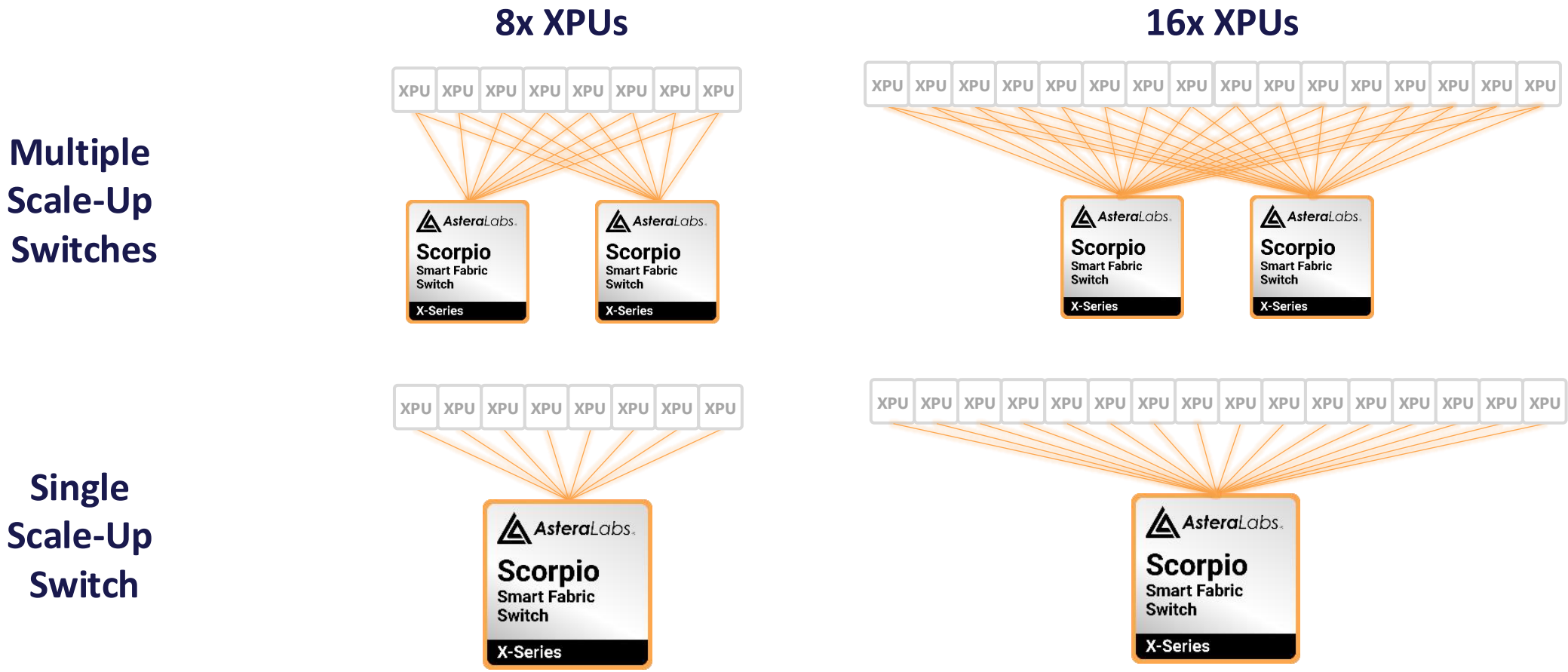
PCIe® Switches for Connecting XPUs, CPUs, NICs, & SSDs





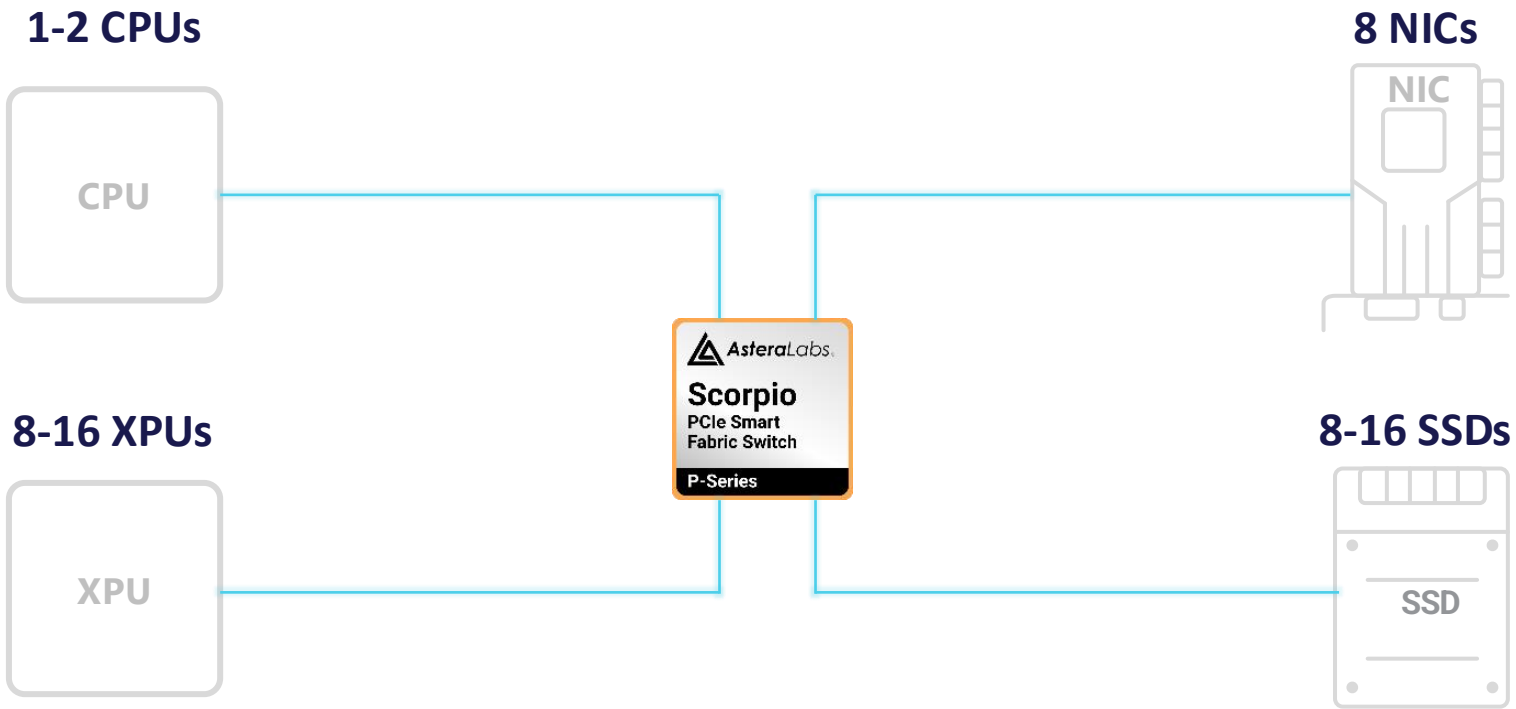
*Several system architectures and topologies enabled by **Scale-Up & Scale-In Switches***

Scale-Up Switch Fabric Topologies



Multiple topologies enabled depending on platform optimizations

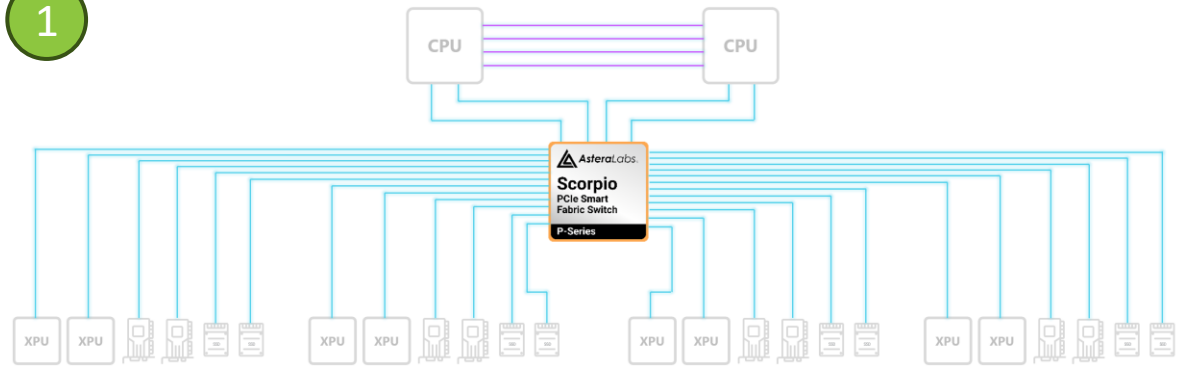
Multiple PCIe® Devices & Limited Ports



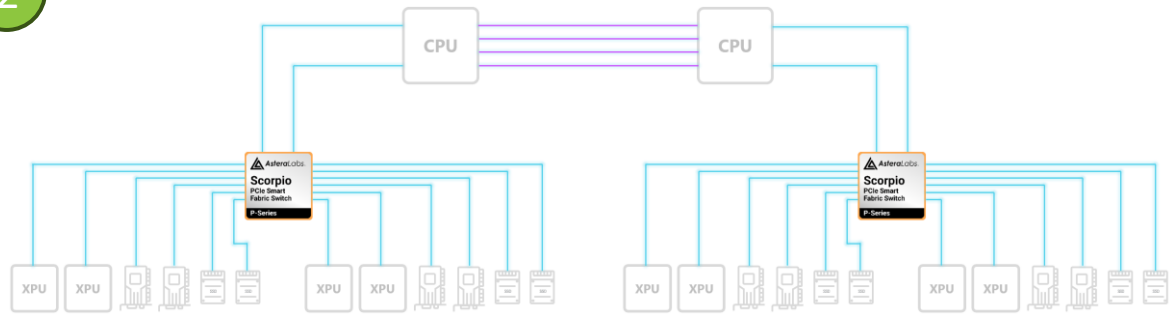
PCIe® switches required to achieve necessary device quantities

PCIe® Switch Fabrics for XPUs, CPUs, NICs, & SSDs

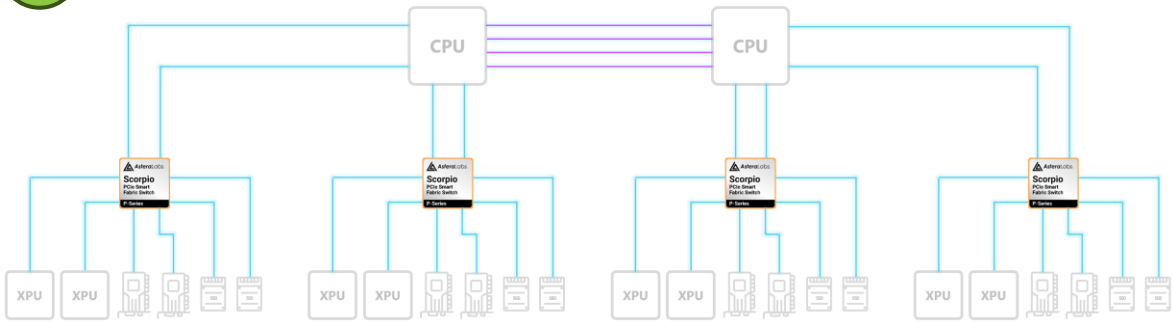
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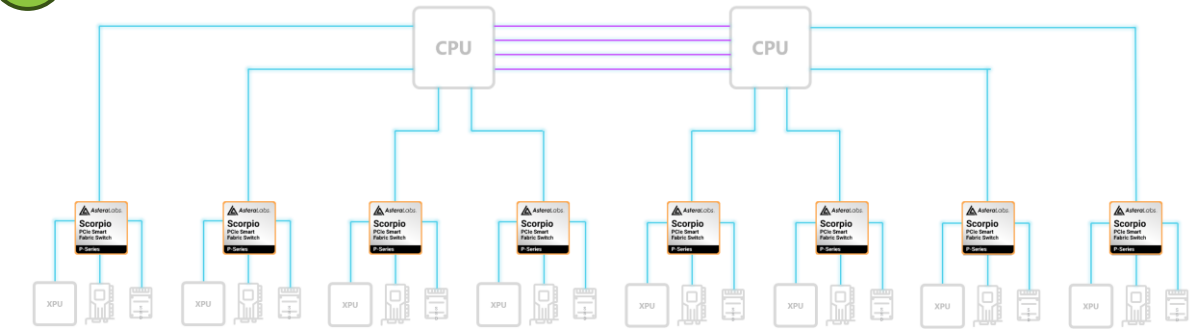
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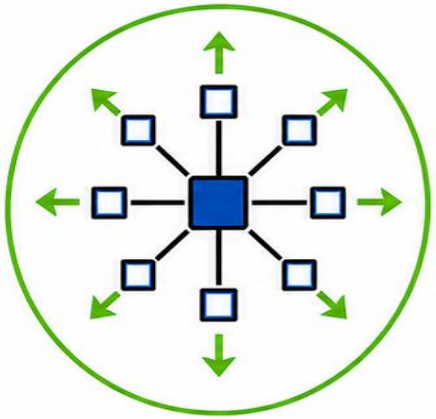


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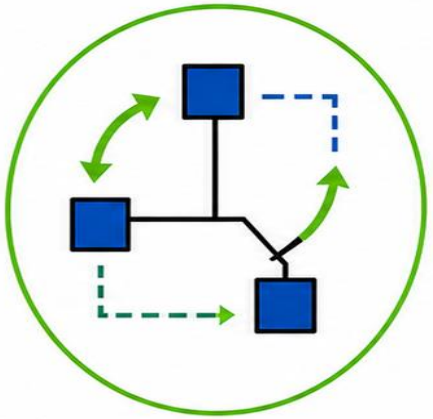
Multiple topologies enabled depending on platform optimizations

Switches are Critical to the OAC Platform



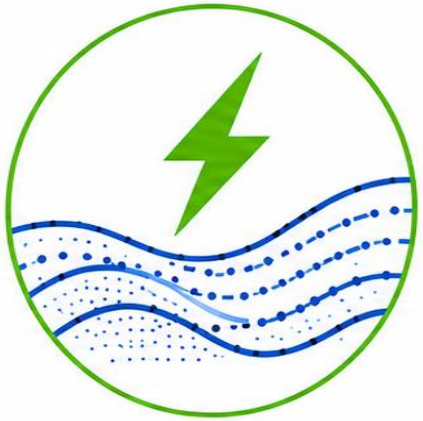
Scalability

Fan-out to more devices



Flexibility

Reconfigure topology without hardware changes



Performance

Non-blocking, high-bandwidth, Low-latency fabric



Manageability

Centralized topology visibility and control



Openness

Open standards vs. Proprietary solutions

Enabling Switch Features & Attributes

- High radix scale-up switches for optimized XPU/GPU fabric
- Various port and lane count options for PCIe® switches to match various OAC system topologies
- Low power and small footprint for switching on OAC modules
- Advanced telemetry and manageability (link health, bandwidth monitoring)
- Key differentiators: *Performance, power efficiency, software ecosystem, and OCP ecosystem engagement*

Summary & Key Takeaways

- OAC is an open, modular AI accelerator architecture with a rich ecosystem of silicon and peripherals
- Switches operate at **two distinct and critical areas** of OAC — scale-up fabric, and PCIe® switch fabric for XPUs, CPUs, NICs, and SSDs
- Without switches, OAC complexity is unmanageable — with switches, it becomes elegant and scalable
- Open switch standards (OCP, PCIe, UALink, ESUN) are key to breaking vendor lock-in in AI infrastructure

Call to Action

- Get involved in the OAC working group
  - Can we drive OAC to true interoperability like we have in a PCIe AIC?
  - Can we align on reference designs?

- Visit the **Astera Labs Booth B34** to learn more about the switches that enable these architectures

- Visit **www.asteralabs.com** for more information

- Come chat with me or drop me a line: **Christopher.Blackburn@asteralabs.com**

Thank You!

